‘ChromoBot’

A Smart Color Sorting & Obstacle Avoiding Line Follower

HND IN SOFTWARE ENGINEERING 25.2F

PROJECT PROPOSAL

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(A) Define the Problem

Modern warehouses need fast, accurate, and safe movement of items. Human workers spend a lot of time manually sorting objects and moving them between areas. This slows down operations and increases the risk of errors, fatigue, and accidents. To address these challenges, automation through robotics offers a solution that improves efficiency, accuracy, and safety.

1.Identifying the Purpose of the Construction

Purpose of the Robot

The **Smart Warehouse Sorting Robot** aims to perform:

- **Automatic color-based item sorting** Detects object colors using a sensor and sorts them into designated bins.
- **Line following for accurate path navigation** Uses IR sensors to follow a predefined path for consistent movement.
- **Obstacle avoidance to prevent collisions** Employs an ultrasonic sensor to detect and avoid obstacles safely.
- **Safe and efficient transportation inside warehouses** Provides stable 4-wheel drive locomotion for reliable movement of items.

This robot helps **automate repetitive tasks** while reducing human effort, minimizing errors, and increasing accuracy in warehouse operations.

2. Identifying Specific Requirements

The robot must:

- **Detect different item colors** using a TCS3200/TCS230 color sensor for sorting decisions.
- **Follow a predefined black line** on a contrasting white surface using IR line sensors.
- **Avoid obstacles** in its path using ultrasonic sensors (HC-SR04).
- **Move using four DC geared motors** (6V–12V, same RPM) controlled by dual L298N motor drivers (or a high-current driver like TB6612FNG).
- **Operate on Arduino Uno**, which serves as the main control unit for sensors and actuators.
- **Transport small objects** using a servo-controlled gripper mechanism (or platform, depending on design).

(B) Researching & Designing

1. Gathering Information & Feasibility Study

Research insights:

- Line-following robots are easy to implement using IR sensors.
- Color sensors provide accurate RGB frequency data, suitable for sorting.
- Arduino UNO supports nearly real-time sensor reading and motor control.
- Obstacle avoidance using HC-SR04 ultrasonic sensor is reliable in indoor spaces.

Feasibility:

- All components are affordable, available in Sri Lanka, and compatible.
- The robot can be built on a **4-wheel chassis** with four DC geared motors for stability.
- Dual motor drivers (L298N × 2 or TB6612FNG) can handle the current requirements.

2. Identifying Design Requirements

The design must include:

- A stable **4-wheel chassis**.
- Correct sensor placements for line detection, obstacle avoidance, and color sorting.
- Proper wiring and power distribution with regulated voltage.
- Code that handles:
  - Line following
  - Color sorting decision
  - Avoiding obstacles
- Battery safety and voltage regulation.
- Easy maintenance and modular connections.

3. Possible & Alternative Solutions

Feature	Primary Solution	Alternative Solution
Line Following	IR Sensor Array	Single IR sensor
Obstacle Avoidance	HC-SR04 ultrasonic	IR proximity sensor
Sorting System	Color sensor + servo gate	Weight-based sorting
Control	Arduino UNO	ESP32 or Arduino Nano

3. Planning & Designing the Structure

Mechanical Design:

- **Four-wheel drive base** with two motors on each side (left/right).
- IR sensor module mounted at the bottom front for line detection.
- Ultrasonic sensor mounted at the top front for obstacle detection.
- Color sensor positioned above the gripper/object tray for accurate scanning.
- Servo-controlled gripper mechanism for picking and placing objects.
- Small containers or bins for sorted items mounted at designated positions.

(C) Creating a Prototype

1.Prototype Development Activities:

- A working simulation was created in Tinkercad for testing connections.
- Individual modules like IR sensors, ultrasonic sensors, and motors were tested.
- Color sensor accuracy was checked by reading RGB values.
- Sorting mechanism tested using a servo motor and gripper.

2.Prototype Troubleshooting:

- IR sensors adjusted for correct height and sensitivity.
- Ultrasonic sensor calibrated to avoid false readings.
- Color sensor tuned for different lighting conditions.
- Motor driver connections reinforced for stability with 4 motors.

(D) Building the Robot

- Mounting Arduino, motor drivers, and sensors on the chassis.
- Connecting motors to wheels and securing them with screws.
- Installing the gripper mechanism with servo motors.
- Wiring all components neatly using cable ties and spacers.
- Integrating the battery pack with a power switch for safe operation.

(E) Evaluating the Robot

1.Evaluating the Design

The robot successfully performed line following, obstacle avoidance, and color sorting. The 4-wheel design provided stability, and the servo-based gripper worked reliably.

2.Evaluating the Planning Process

Subsystems were researched and tested before integration, which reduced errors. The chosen design balanced simplicity and functionality, and the modular approach made troubleshooting straightforward.